

Project Plan and Paper Structure

How Stable Are Regression Conclusions Across Subpopulations? A Finite-Sample Empirical Study

Draft planning document

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1 Working Title

How Stable Are Regression Conclusions Across Subpopulations? A Finite-Sample Empirical Study

2 Short Answer: Is a Literature Review Necessary?

Yes, but only a **small and focused** one.

For this project, a long standalone literature review is probably unnecessary. A short review embedded in the introduction is enough if it does the following:

- (1) explains why pooled regression across heterogeneous groups can be misleading,
- (2) briefly connects the paper to existing work on subgroup heterogeneity, model transportability, and distribution shift,
- (3) identifies the gap your paper addresses,
- (4) clarifies your empirical contribution.

A good target is roughly **3–6 paragraphs** in the introduction devoted to prior work and positioning.

3 Project Goal

The goal of this paper is to study how stable the conclusions of linear regression are across subpopulations in finite samples.

More precisely, the paper asks:

When a linear regression model is fit on pooled data or on one subgroup, how stable are the resulting conclusions—such as coefficient signs, coefficient magnitudes, statistical significance, and predictive performance—when transferred to other subgroups?

The project will combine:

- a controlled **simulation study**, and
- a small number of **real-data case studies**.

4 Core Research Questions

The paper will focus on the following questions:

- RQ1.** Do coefficient **signs** remain stable across subpopulations?
- RQ2.** How much do coefficient **magnitudes** drift across subpopulations?
- RQ3.** When does a **pooled model** hide subgroup-specific relationships?
- RQ4.** To what extent can **interaction models** recover subgroup heterogeneity?
- RQ5.** How different is **predictive performance across groups** when a model is transported from one subgroup to another?

5 Main Working Hypothesis

The main hypothesis is:

Even when a pooled linear regression appears adequate in aggregate, the regression conclusions may be unstable across subpopulations due to subgroup heterogeneity in coefficient values, predictor distributions, and noise structure.

A secondary hypothesis is:

Predictive performance may remain moderately stable even when coefficient interpretation is not, so predictive adequacy alone does not guarantee interpretive transportability.

6 Main Contribution

This paper is not intended to propose a new regression estimator. Instead, its contribution is primarily empirical and methodological.

The intended contributions are:

- (1) a finite-sample empirical study of regression instability across subpopulations,
- (2) a comparison of pooled, subgroup-specific, and interaction-based models,
- (3) a small set of interpretable stability metrics,
- (4) practical guidance on when pooled regression conclusions should be treated with caution.

7 Proposed Stability Metrics

The study will use a small number of simple and interpretable metrics.

7.1 Sign Stability

For each predictor, compare the sign of the estimated coefficient across subgroups and with the pooled estimate.

A simple summary metric can be:

$$\text{Sign Stability Rate}_j = \frac{\#\{\text{comparisons where the sign agrees}\}}{\#\{\text{all relevant comparisons}\}}.$$

7.2 Standardized Coefficient Drift

For predictor j , compare subgroup-specific estimates with the pooled estimate:

$$D_j = \max_g \left| \hat{\beta}_{j,g} - \hat{\beta}_{j,\text{pooled}} \right|.$$

Report the standardized drift

$$D_j^* = \frac{D_j}{\widehat{\text{SE}}(\hat{\beta}_{j,\text{pooled}})}$$

to make comparisons across predictors more interpretable.

7.3 Transported Prediction Gap

Train a model on one subgroup and evaluate it on another subgroup. Report metrics such as RMSE or MAE. The gap in out-of-group performance will quantify loss of transportability.

7.4 Covariate Shift Summary

Quantify predictor distribution differences across groups using a compact summary, such as the average standardized mean difference across predictors. This provides context for when cross-group transport failures are likely.

8 Models to Compare

The paper will compare the following models:

Model 1: Pooled OLS

$$Y = \beta_0 + X^\top \beta + \varepsilon.$$

Model 2: Pooled OLS with Group Indicator

$$Y = \beta_0 + X^\top \beta + \gamma G + \varepsilon.$$

Model 3: Pooled OLS with Group Interactions

$$Y = \beta_0 + X^\top \beta + \gamma G + X^\top \delta G + \varepsilon.$$

Model 4: Subgroup-Specific OLS

A separate linear regression will be fit within each subgroup.

Optional extensions such as ridge or lasso may be considered only if time permits, but they are not central to the project.

9 Simulation Study

The simulation study is the core of the paper.

9.1 Basic Setup

Generate data for G subgroups, each with predictors and response

$$X_g \sim \mathcal{N}(\mu_g, \Sigma_g), \quad Y_g = \beta_{0,g} + X_g^\top \beta_g + \varepsilon_g,$$

where

$$\varepsilon_g \sim \mathcal{N}(0, \sigma_g^2).$$

9.2 Simulation Scenarios

The study will consider a small number of scenarios:

- S1. No heterogeneity:** same coefficients, same predictor distribution, same noise across groups.
- S2. Covariate shift only:** same coefficients, different predictor distributions across groups.
- S3. Coefficient heterogeneity only:** different coefficients, same predictor distributions.
- S4. Combined heterogeneity:** different coefficients and different predictor distributions.

If time allows, one additional scenario may be added with different noise levels across subgroups.

9.3 Simulation Factors

The simulation will vary a small set of factors:

- sample size,
- number of predictors,
- degree of collinearity,
- signal-to-noise ratio,
- subgroup balance.

9.4 Simulation Outputs

For each replicate, the study will record:

- estimated coefficients,
- coefficient signs,
- standard errors and confidence intervals,
- predictive performance within and across groups,
- interaction detection behavior.

10 Real-Data Case Studies

The empirical section should remain small and focused. Two or three real datasets are enough.

Desired dataset properties:

- continuous response variable,
- at least one clear subgroup variable,
- multiple predictors,
- manageable missingness and preprocessing requirements,
- interpretable context.

Examples of subgroup variables include sex, age category, hospital/site, region, cohort, or socioeconomic category.

The purpose of the real-data section is not to establish universal truth, but to illustrate the practical relevance of the simulation findings.

11 Why This Topic Matters

This project matters for at least three reasons:

- (1) In applied work, analysts often fit a single pooled regression and interpret the coefficients as if they represented one stable relationship.
- (2) In reality, subgroup heterogeneity may make such interpretations unreliable, even when the overall fit looks acceptable.
- (3) In many practical settings, predictive performance alone is not enough; analysts also care about whether the sign and size of an effect are stable across contexts.

12 Mini Literature Review

This review should be short and functional, not exhaustive.

12.1 Suggested Themes to Cover

The literature discussion in the introduction should briefly cover the following themes:

- (1) **Classical regression and heterogeneity.** Standard linear regression often assumes a common linear relationship across observations unless interactions or stratification are introduced.
- (2) **Subgroup effects and interaction modeling.** It is well known that subgroup-specific relationships may be masked in pooled analyses, and interaction terms are a natural way to represent effect heterogeneity.
- (3) **Transportability and distribution shift.** Modern statistics and machine learning increasingly study how model conclusions and predictive performance change under population shift or domain shift.

- (4) **Interpretation versus prediction.** A model may predict reasonably well across groups while still producing unstable or misleading coefficient interpretations.
- (5) **Gap addressed by this paper.** Much of the existing literature is either broad and theoretical, or focused on general distribution shift. This paper instead gives a narrow finite-sample empirical study centered specifically on the stability of regression conclusions across subpopulations.

12.2 What the Literature Review Should *Not* Try to Do

Do not try to:

- survey all of domain adaptation or all of distribution shift,
- review all heterogeneity modeling methods,
- make the paper look theoretical if it is mainly empirical,
- overclaim novelty by pretending no related work exists.

13 Proposed Paper Structure

1. Introduction

This section should:

- motivate the problem,
- explain why pooled regression may be misleading across subpopulations,
- provide a short literature review,
- identify the paper’s gap,
- state the paper’s contributions.

2. Methods

This section should define:

- notation,
- subgroup setting,
- models being compared,
- stability metrics,
- simulation design,
- evaluation protocol for real data.

3. Simulation Study

This section should present:

- scenario definitions,
- simulation setup,
- main figures and tables,
- interpretation of the observed instability patterns.

4. Real-Data Case Studies

This section should:

- describe the selected datasets,
- define subgroup variables,
- compare pooled and subgroup-aware models,
- connect the findings back to the simulation results.

5. Discussion

This section should:

- summarize the main findings,
- explain practical implications,
- discuss limitations,
- suggest future extensions.

6. Conclusion

This should be short and direct.

14 Practical Goals for the Project

The project should aim to produce the following:

- G1.** A reproducible simulation pipeline in R.
- G2.** A clear set of 3–5 main figures.
- G3.** A small number of interpretable summary tables.
- G4.** Two strong real-data illustrations.
- G5.** A paper draft that is concise, coherent, and clearly positioned.

15 What to Prioritize

If time becomes limited, prioritize in this order:

- (1) simulation study,
- (2) clear metrics,
- (3) good figures,
- (4) concise writing,
- (5) real-data illustrations,
- (6) optional extensions.

16 What to Avoid

To keep the project publishable and finishable, avoid:

- too many models,
- too many datasets,
- unnecessary deep learning baselines,
- excessive theory,
- broad claims about causality,
- trying to solve all of distribution shift.

17 Minimal Viable Version of the Paper

If the project must be compressed, the minimum viable paper is:

- one clear introduction with a short literature review,
- one methods section,
- one simulation section with 4 scenarios,
- one real-data section with 2 datasets,
- one discussion/conclusion section.

18 Final Recommendation

Yes, include a literature review, but make it **short, focused, and embedded in the introduction**. For this paper, the strongest version is not one with the longest review, but one with:

- a precise question,
- a clean simulation design,
- strong figures,
- a credible gap statement,
- and disciplined scope.